

Lec 3: algebraic groups & Lie algebras III

1) Distribution algebras

1.0) Introduction.

Let F be an algebraically closed field, $G \subset GL_n(F)$ algebraic subgroup & $\mathfrak{g} = T_e G \subset Mat_n(F)$, the tangent space at the unit e . We want to understand additional structures on $\mathfrak{g}$. In Lec 2, Sec 2, we stated:

Thm: 1) $\mathfrak{g}$ is closed under $[\cdot, \cdot]: \xi, \eta \in \mathfrak{g} \Rightarrow [\xi, \eta] = \xi\eta - \eta\xi \in \mathfrak{g}$.

1') If $\text{char } F = p > 0$, then $\mathfrak{g}$ is also closed under $\xi \mapsto \xi^p$.

Let $H \subset GL_m$ be another algebraic subgroup & $\varphi: G \rightarrow H$ be an algebraic group homomorphism & $\varphi = T_e \varphi$. Then

2) $[\varphi(\xi), \varphi(\eta)] = \varphi([\xi, \eta]) \quad \forall \xi, \eta \in \mathfrak{g}$.

2') If $\text{char } F = p > 0$, then $\varphi(\xi^p) = \varphi(\xi)^p \quad \forall \xi \in \mathfrak{g}$.

Idea of proof: $\bullet (\mathcal{D}(G), *)$ - distributions supported at e .

$\varphi: G \rightarrow H \leadsto$ algebra homom. $\mathcal{D}(G) \rightarrow \mathcal{D}(H)$

$\mathfrak{g} \subset \mathcal{D}(G)$ as subspace.

- $\mathfrak{gl}_n \subset \mathcal{D}(\mathbb{A}^n)$: $\xi * \eta - \eta * \xi = [\xi, \eta] \in \mathfrak{gl}_n \quad \forall \xi, \eta \in \mathfrak{gl}_n$
 $\xi^{*p} = \xi^p \in \mathfrak{gl}_n$.
- finish the proof.

1.1) Space of distributions.

X an affine variety, $A := \mathbb{F}[X]$, $\alpha \in X \leadsto \mathfrak{m}_\alpha = \{f \in A \mid f(\alpha) = 0\}$ max. ideal

A distribution on X supported at α is any $\delta \in A^*$ |

$$\exists k \geq 0 \mid \delta(\mathfrak{m}_\alpha^k) = 0 \iff \delta \in (A/\mathfrak{m}_\alpha^k)^* \subset A^*$$

Subspace of distributions $\mathcal{D}_\alpha(X) = \bigcup_{k \geq 0} \underbrace{(A/\mathfrak{m}_\alpha^k)^*}_{\text{ascending chain}} \subset A^*$

Example: $X = \mathbb{F}$, $\alpha = 0$. $A = \mathbb{F}[x]$, $\mathfrak{m}_\alpha = (x)$, $A/\mathfrak{m}_\alpha^k = \text{Span}(1, \dots, x^{k-1})$

$\mathcal{D}_\alpha(X)$ has basis $\delta^{(i)}$, $i \geq 0$, $\delta^{(i)}(x^j) = \delta_{ij}$ (Kronecker δ)

$\varphi: X \rightarrow Y$ morphism $\leadsto$ homomorphism $\varphi^*: \mathbb{F}[Y] \rightarrow \mathbb{F}[X] \leadsto$
 $\varphi_*: \mathbb{F}[X]^* \rightarrow \mathbb{F}[Y]^*$.

Lemma: $\varphi_*(\mathcal{D}_\alpha(X)) \subset \mathcal{D}_{\varphi(\alpha)}(Y)$.

Proof:

By Lemma in Sec 1.2 in Lec 2 (its proof):

$$\varphi^*(\mathfrak{m}_{\varphi(\alpha)}^k) \subset \mathfrak{m}_\alpha^k \Rightarrow \text{if } \delta(\mathfrak{m}_\alpha^k) = 0 \text{ \& } f \in \mathfrak{m}_{\varphi(\alpha)}^k \Rightarrow$$

$$\varphi_* \delta(f) = \delta(\varphi^*(f)) = 0 \quad \square$$

Remark: $T_\alpha X = \{ \zeta \in A^* \mid \zeta(1) = \zeta(\mathfrak{m}_\alpha^2) = 0 \} \subset \mathcal{D}_\alpha(X)$

1.2) Group case

$X := G$ (affine alg. group), $\alpha = e \leadsto \mathcal{D}(G) := \mathcal{D}_e(G)$

Map: $\ast: \mathcal{D}(G) \times \mathcal{D}(G) \rightarrow \mathcal{D}(G)$

product $\uparrow$
 $\mu: G \times G \rightarrow G \leadsto \mu^*: \mathbb{F}[G] \rightarrow \mathbb{F}[G \times G] =$
 $\mathbb{F}[G] \otimes \mathbb{F}[G]$

$f \in \mathbb{F}[G], \delta_1, \delta_2 \in \mathcal{D}(G) \leadsto [\delta_1 \ast \delta_2](f) := [\delta_1 \otimes \delta_2](\mu^*(f))$

where $\delta_1 \otimes \delta_2: \mathbb{F}[G] \otimes \mathbb{F}[G] \rightarrow \mathbb{F}, f_1 \otimes f_2 \mapsto \delta_1(f_1) \delta_2(f_2)$

So $\delta_1 \ast \delta_2 \in \mathbb{F}[G]^*$

Example cont'd: $G = \mathbb{G}_a (= (\mathbb{F}, +)), \alpha = 0 \leadsto \mathcal{D}(G) = \text{Span}(\delta^{(i)})$

Want: $\delta^{(i)} \ast \delta^{(j)}$; $\mu(a_1, a_2) = a_1 + a_2, \mathbb{F}[G] = \mathbb{F}[x],$

$\mathbb{F}[G \times G] = \mathbb{F}[x_1, x_2], x_1 = x \otimes 1, x_2 = 1 \otimes x \Rightarrow \mu^*(x) = x_1 + x_2$

$$\begin{aligned} \delta^{(i)} * \delta^{(j)}(x^k) &= [\delta^{(i)} \otimes \delta^{(j)}](\mu^*(x^k) = (x_1 + x_2)^k) = [\text{coeff. of } \\ x^i \otimes x^j = x_1^i x_2^j \text{ in } (x_1 + x_2)^k] &= \left[\binom{i+j}{i} \delta^{(i+j)} \right](x^k) \Rightarrow \\ \delta^{(i)} * \delta^{(j)} &= \binom{i+j}{i} \delta^{(i+j)} \end{aligned}$$

Interpretation: $\text{Span}_{\mathbb{Z}} \left(\underbrace{\frac{\delta^i}{i!}}_{B^i} \mid i \geq 0 \right) \subset Q[\delta] \text{-subring}$

$$\sim \mathbb{F} \otimes_{\mathbb{Z}} B \xrightarrow{\sim} \mathcal{D}(\mathbb{G}_a), \quad 1 \otimes \frac{\delta^i}{i!} \leftrightarrow \delta^{(i)}$$

If $\text{char } \mathbb{F} = 0 \Rightarrow \mathcal{D}(\mathbb{G}_a) = Q[\delta]$

$\text{char } \mathbb{F} = p \Rightarrow \mathcal{D}(\mathbb{G}_a) \text{ not fin. gen'd.}$

$$\mathfrak{g} = T_0 \mathbb{G}_a = \mathbb{F} \subset \mathcal{D}(\mathbb{G}_a) \text{ as } \mathbb{F} \delta^{(1)}$$

Premium example/exercise: $\mathcal{D}(\mathbb{G}_m)$

$$C = \text{Span}_{\mathbb{Z}} \left(\underbrace{\binom{\delta}{i}}_{\substack{\delta(\delta-1)\dots(\delta-i+1) \\ i!}} \mid i \geq 0 \right) \subset Q[\delta] \text{-subring.}$$

$$\begin{aligned} \mathcal{D}(\mathbb{G}_m) &\xrightarrow{\sim} \mathbb{F} \otimes_{\mathbb{Z}} C \quad (\text{hint: } \delta_i \in \mathcal{D}(\mathbb{G}_m), \delta_i((x-1)^j) = \delta_{ij}) \\ \delta_i &\leftrightarrow 1 \otimes \binom{\delta}{i} \end{aligned}$$

Proposition: 0) $\mathcal{D}(G) * \mathcal{D}(G) \subset \mathcal{D}(G)$

1) $(\mathcal{D}(G), *)$ is unital associative algebra (unit ε :

$$\varepsilon(1) = 1, \varepsilon(m_e) = 0$$

2) Alg. group homom. $\varphi: G \rightarrow H \Rightarrow \varphi_*: D(G) \rightarrow D(H)$ is unital algebra homomorphism.

Proof:

$$0) A = \mathbb{F}[G], m = m_e \subset A, \delta_i \in (A/m^{k_i})^* \quad k_i \in \mathbb{Z}_{\geq 0}, i=1,2.$$

$$\text{Claim: } [\delta_1 * \delta_2](m^{k_1+k_2}) = 0$$

$$[\delta_1 * \delta_2](f) = [\delta_1 \otimes \delta_2](\mu^*(f))$$

$$0 \text{ on } \underbrace{m^{k_1} \otimes A + A \otimes m^{k_2}} \subset A \otimes A$$

$$\text{WTS: } f \in m^{k_1+k_2} \Rightarrow \mu^*(f)$$

$$\underbrace{m_{(e,e)}}_{m \otimes A + A \otimes m} \subset \mathbb{F}[G] \otimes \mathbb{F}[G] \text{-max. ideal; } \mu(e,e) = e \Rightarrow$$

Exercise:

$$m_{(e,e)} \supset m \otimes A + A \otimes m$$

$$\text{for } \subset: A \otimes A = \mathbb{F}(1 \otimes 1) + m \otimes A + A \otimes m$$

$$\text{check: } f_1 \otimes f_2 \in \overset{\uparrow}{\text{r.h.s.}}$$

$$+ \mu^*(m) \subset m_{(e,e)} = m \otimes A + A \otimes m \Rightarrow \mu^*(m^{k_1+k_2}) \subset \mu^*(m)^{k_1+k_2}$$

$$\subset (m \otimes A + A \otimes m)^{k_1+k_2} = [\text{binomial}] \subset m^{k_1} \otimes A + A \otimes m^{k_2} \quad \square \text{ of 0)$$

1): $*$ is associative $\Leftarrow \mu$ is associative

$\Downarrow$ the following is commut.

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{\mu \times \text{id}} & G \times G \\ \downarrow \text{id} \times \mu & & \downarrow \mu \\ G \times G & \xrightarrow{\mu} & G \end{array}$$

$\Downarrow$ diagram of pullbacks is commut.

$$\begin{array}{ccc} A \otimes A \otimes A & \xleftarrow{(\mu \times \text{id})^* = \mu^* \otimes \text{id}} & A \otimes A \\ \uparrow (\text{id} \times \mu)^* = \text{id}_A \otimes \mu^* & & \uparrow \mu^* \\ A \otimes A & \xleftarrow{\mu^*} & A \end{array}$$

$a \otimes b \mapsto \mu^*(a) \otimes b$

$$[(\delta_1 * \delta_2) * \delta_3](f) = [(\delta_1 * \delta_2) \otimes \delta_3](\mu^*(f))$$

$$[\delta_1 \otimes \delta_2 \otimes \delta_3](\underbrace{[\mu^* \otimes \text{id}] \circ \mu^*}_{\text{pullback}}(f))$$

$$[\delta_1 * (\delta_2 * \delta_3)](f) = (\text{id} \otimes \mu^*) \circ \mu^*(f)$$

ε is unit (exercise).

$\square$ of 1)

2) Similar to 1) but using commut. diagram

$$\begin{array}{ccc}
 G \times G & \xrightarrow{\mu_G} & G \\
 \downarrow \varphi \times \varphi & & \downarrow \varphi \\
 H \times H & \xrightarrow{\mu_H} & H
 \end{array}$$

Proof - see posted notes

□

1.3) Case of GL_n

Proposition: $\bullet A, B \in \mathfrak{gl}_n \subset \mathcal{D}(GL_n) \Rightarrow A*B - B*A = [A, B] \in \mathfrak{gl}_n$

$\bullet \text{char } F = p \Rightarrow A^{*p} = A^p \in \mathfrak{gl}_n$.

Proof (Taylor series)

• Commutator

$$A \in T_e GL_n: A.f = \frac{1}{t} (f(I + At) - f(I))|_{t=0}$$

$$= [\text{coeff. of } t \text{ in Taylor series of } f(I + At)]$$

$$F \in F[GL_n \times GL_n]$$

$$(1) [A \otimes B](F) = [\text{coeff. of } ts \text{ in } F(I + tA, I + sB)]$$

↑
true for $F = f_1 \otimes f_2$

$$F = \mu^*(f), \mu^*(f)(I + tA, I + sB) = f((I + tA)(I + sB))$$

$$= f(I + tA + sB + tsAB)$$

(1) is sum of:

(i) Contribution of tA, tB :

$$\left[\sum_{i,j,k,e=1}^n \frac{\partial^2 f}{\partial x_{ij} \partial x_{ke}} (I) A_{ij} B_{ke} \right]$$

$\nwarrow$ $\nearrow$ $\nwarrow$ $\nearrow$
 corresp. function (i,j) -matrix entry
 symmetric w.r.t. $A \& B$

(ii) Contribution of tAB

$$\sum_{i,j=1}^n \frac{\partial f}{\partial x_{ij}} (I) \cdot (AB)_{ij} = AB \in \text{Mat}_n(F) = T_e GL_n$$

$$[A*B - B*A](f) = [(i) \text{ cancel!}] = AB - BA$$

$\square$ of commutator.

• Case of p th power

$A^{*p}(f) = \text{coeff. of } t_1 \dots t_p \text{ in Taylor expansion of}$

$$(2) f\left(\prod_{i=1}^p (I + t_i A)\right) = f\left(\sum_{S \subset \{1, \dots, p\}} t_S A^{|S|}\right), \quad t_S = \prod_{i \in S} t_i$$

Coeff. t equals:

$$\sum_{\text{over}} \frac{\partial^k f}{\partial x_{i_1 j_1} \dots \partial x_{i_k j_k}} (I) \prod_{e=1}^k (A^{|S_e|})_{i_e j_e}$$

$$\text{Ind} = \{(S_1, \dots, S_k, (i_1, j_1), \dots, (i_k, j_k)) \mid 1 \leq k \leq p, S_1 \cup \dots \cup S_k = \{1, \dots, p\},$$

$$i_e, j_e \in \{1, \dots, n\}$$

Symmetric group $S_p \curvearrowright \{1, \dots, p\} \leadsto S_p \curvearrowright \text{Ind}$, fixing (i_e, j_e) & $|S_e|$

$\Rightarrow$ summands for indexes in the same S_p -orbit are equal

Cardinality of orbit = $\frac{p!}{|S_1|! \dots |S_k|!}$ - divisible by p if $k > 1$
 - the corresponding summands sum to 0.

For $k=1$:

$$\sum_{i,j} \frac{\partial f}{\partial x_{ij}} (A^p)_{ij} = A^p \in \text{Mat}_n(\mathbb{F}) = T_e GL_n \quad \square$$